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(54) **PLASMA PROCESSING APPARATUS AND PLASMA PROCESSING METHOD USING THE SAME**

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None
See application file for complete search history.

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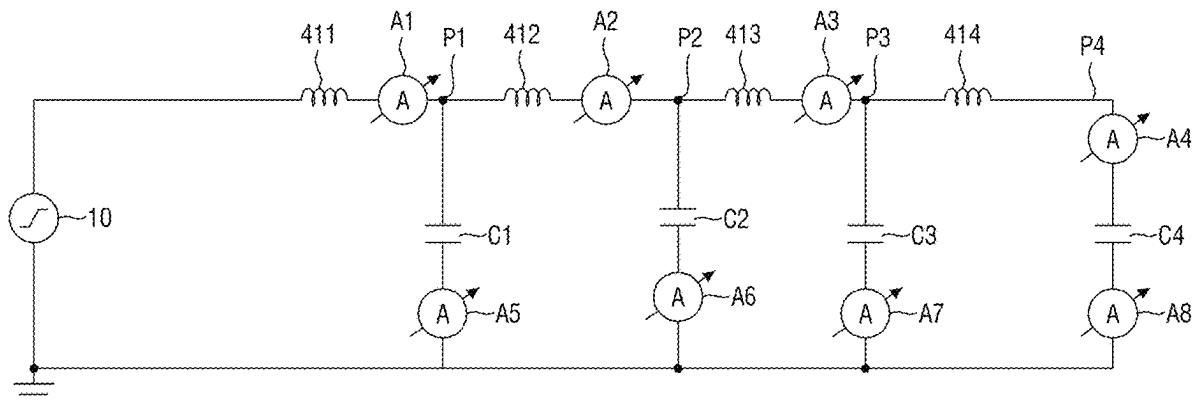
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(57) **ABSTRACT**

A plasma processing apparatus is provided. The plasma processing apparatus includes a chamber having a processing space defined therein in which plasma is generated; and a plasma generation unit configured to excite gas in the processing space into a plasma state, wherein the plasma generation unit includes: a first power supply to supply power for generation of the plasma; a coil connected to the first power supply; a first shunt capacitor disposed between and connected to a first node of the coil and a ground; and a second shunt capacitor disposed between and connected to a second node other than the first node of the coil and the ground.

13 Claims, 5 Drawing Sheets



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FIG. 1

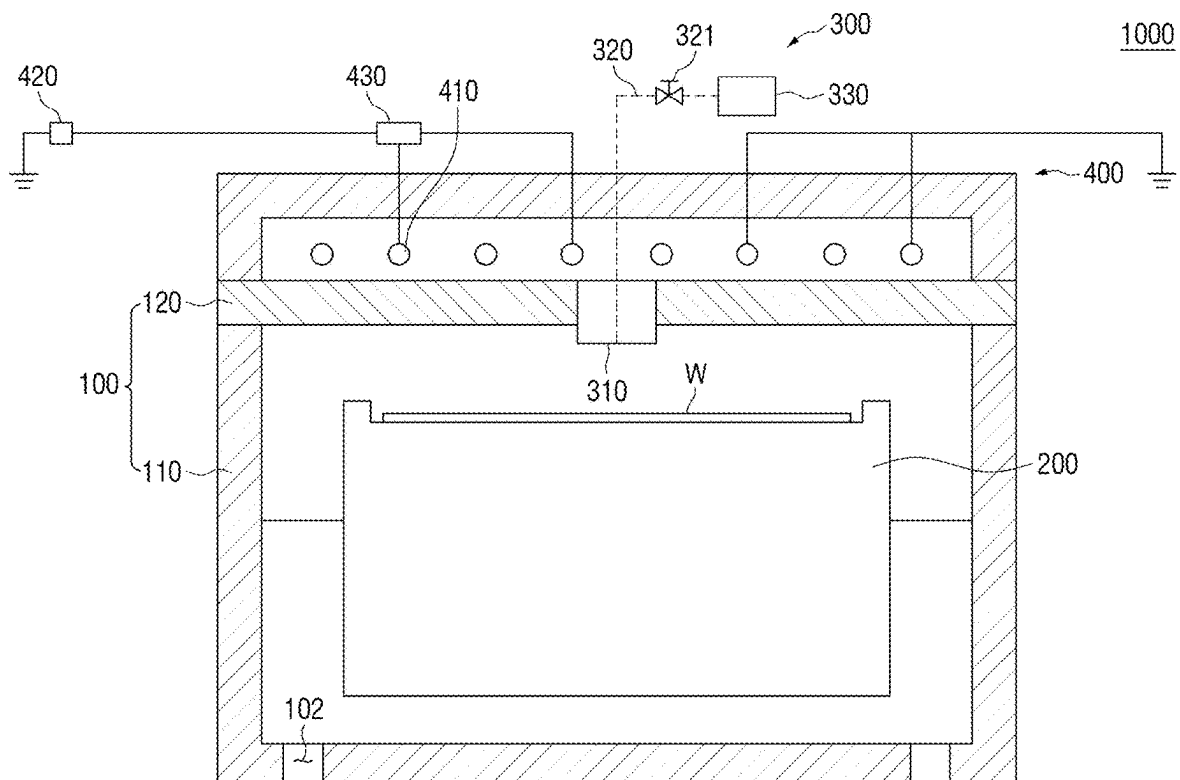


FIG. 2

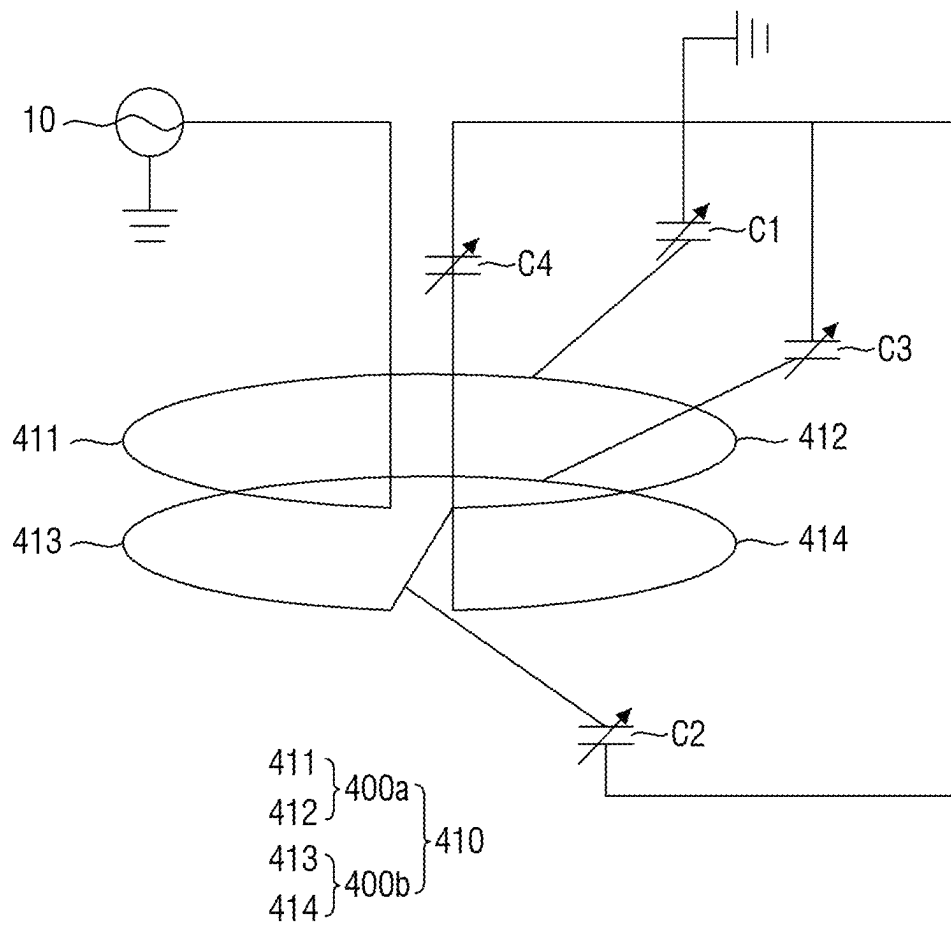


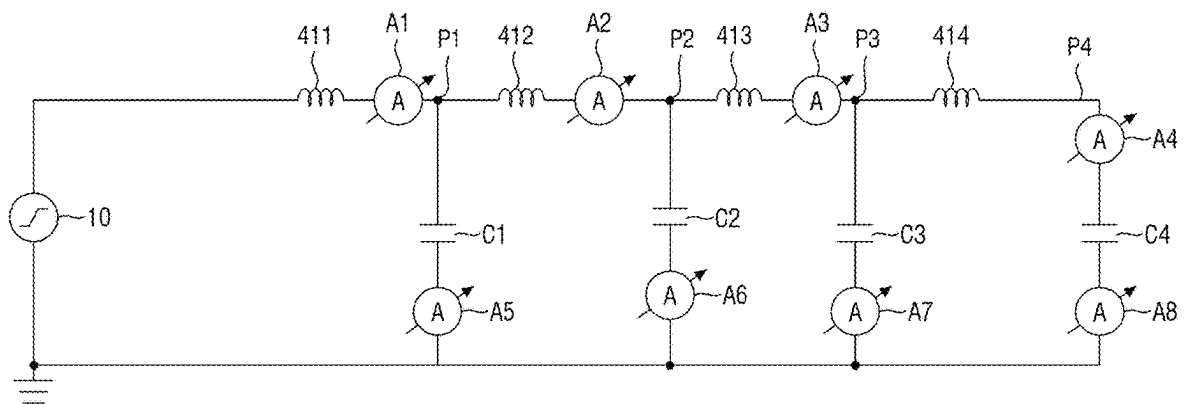
FIG. 3

FIG. 4

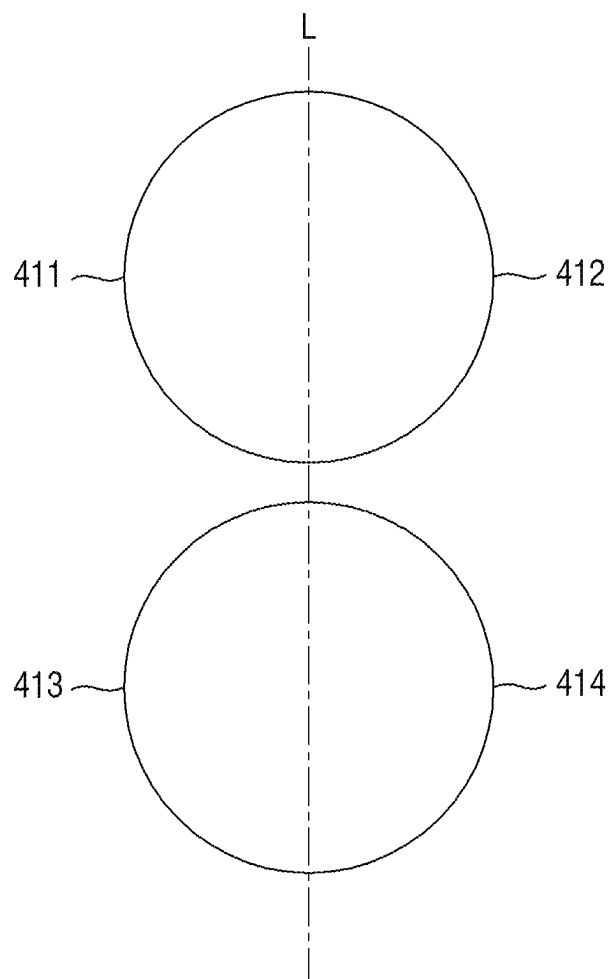
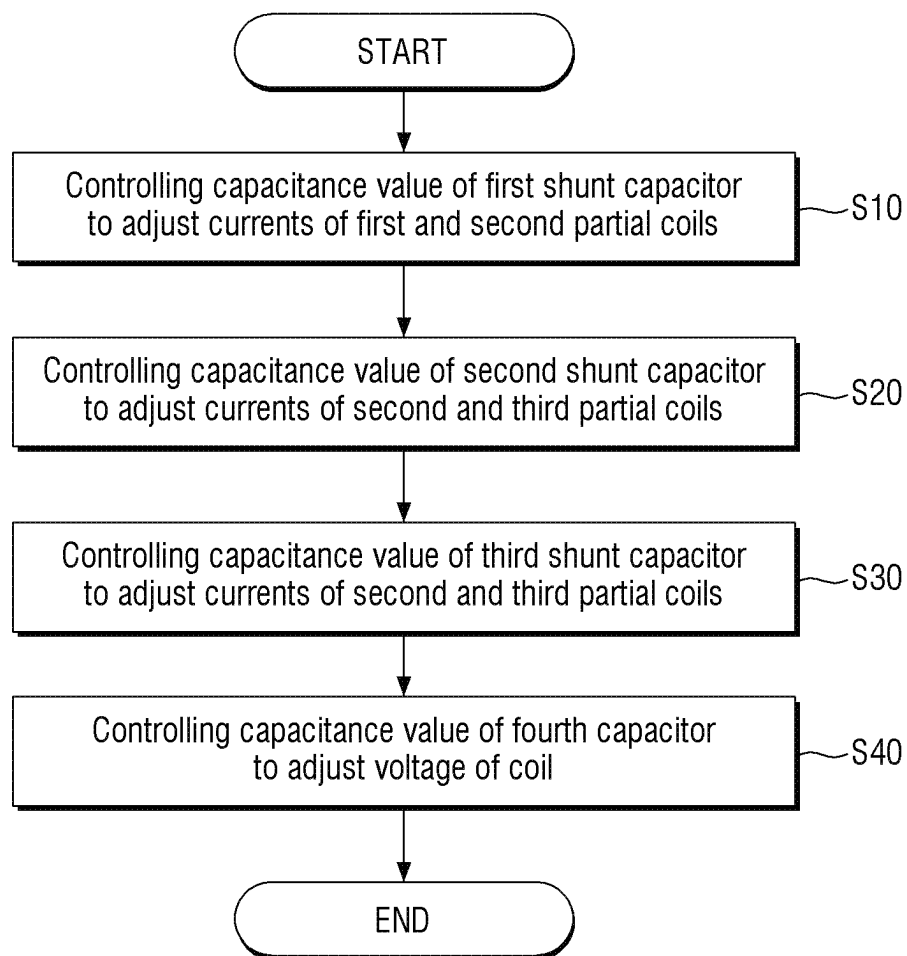


FIG. 5

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PLASMA PROCESSING APPARATUS AND PLASMA PROCESSING METHOD USING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Korean Patent Application No. 10-2021-0139978 filed on Oct. 20, 2021 in the Korean Intellectual Property Office, and all the benefits accruing therefrom under 35 U.S.C. 119, the contents of which in its entirety are herein incorporated by reference.

BACKGROUND

Field

The present disclosure relates to a plasma processing apparatus and a plasma processing method using the same.

Description of Related Art

When manufacturing a semiconductor device or a display device, various processes using plasma, for example, etching, ashing, ion implantation, cleaning, etc. may be used. A substrate processing apparatus using plasma may be classified into CCP (Capacitively Coupled Plasma) type and ICP (Inductively Coupled Plasma) type based on a plasma generation scheme. In the CCP type, two electrodes are disposed to face each other in a chamber, and an RF (radio frequency) signal is applied to one or both of the two electrodes to generate an electric field in the chamber to generate plasma. On the other hand, in the ICP type, one or more coils are installed on the chamber, and the RF signal is applied to the coil to induce an electromagnetic field in the chamber to generate plasma.

SUMMARY

A purpose of the present disclosure is to provide a plasma processing apparatus in which one coil-type antenna is divided into a plurality of portions, and a current of the antenna is adjusted via a shunt capacitor connected to each of the plurality of portions to control a plasma density of each of areas corresponding to the plurality of portions.

A purpose of the present disclosure is to provide a plasma processing method in which one coil-type antenna is divided into a plurality of portions, and a current of the antenna is adjusted via a shunt capacitor connected to each of the plurality of portions to control a plasma density of each of areas corresponding to the plurality of portions.

Purposes according to the present disclosure are not limited to the above-mentioned purpose. Other purposes and advantages according to the present disclosure that are not mentioned may be understood based on following descriptions, and may be more clearly understood based on embodiments according to the present disclosure. Further, it will be easily understood that the purposes and advantages according to the present disclosure may be realized using means shown in the claims and combinations thereof.

A first aspect of the present disclosure provides a plasma processing apparatus comprising: a chamber having a processing space defined therein in which plasma is generated; and a plasma generation unit configured to excite gas in the processing space into a plasma state, wherein the plasma generation unit includes: a first power supply to supply power for generation of the plasma; a coil connected to the

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first power supply; a first shunt capacitor disposed between and connected to a first node of the coil and a ground; and a second shunt capacitor disposed between and connected to a second node other than the first node of the coil and the ground.

In one implementation of the first aspect, the apparatus is configured to control a capacitance value of each of the first and second shunt capacitors to adjust a current of the coil.

In one implementation of the first aspect, the apparatus is configured to adjust a current of the coil to control a density of the plasma.

In one implementation of the first aspect, each of the first and second shunt capacitors includes a variable capacitor.

In one implementation of the first aspect, the coil is divided into first and second partial coils connected in series to each other while the first node is interposed therebetween, wherein the apparatus is configured to control a capacitance value of the first shunt capacitor to adjust currents of the first and second partial coils.

In one implementation of the first aspect, the coil is further divided into a third partial coil connected in series with the second partial coil while the second node is interposed therebetween, wherein the apparatus is configured to control a capacitance value of the second shunt capacitor to adjust currents of the second and third partial coils.

In one implementation of the first aspect, the plasma generation unit further includes a third shunt capacitor disposed between and connected to a third node other than the second node of the coil and the ground, wherein the coil is further divided into a fourth partial coil connected in series with the third partial coil while the third node is interposed therebetween, wherein the apparatus is configured to control a capacitance value of the third shunt capacitor to adjust currents of the third and fourth partial coils.

In one implementation of the first aspect, the plasma generation unit further includes a fourth capacitor disposed between and connected to a fourth node other than the third node of the coil and the ground, wherein the apparatus is configured to control a capacitance value of the fourth capacitor to adjust a voltage of the coil.

In one implementation of the first aspect, the coil includes: a first coil portion having one end connected to the first power supply, wherein the first coil portion is divided into first and second areas; and a second coil portion disposed below the first coil portion and connected to the other end of the first coil portion, wherein the second coil portion is divided into third and fourth areas, wherein each of the first area and the third area constitutes one side half of each of the first and second coil portions while the first area and the third area face vertically each other, wherein each of the second area and the fourth area constitutes the other side half of each of the first and second coil portions while the second area and the fourth area face vertically each other, wherein the apparatus is configured to control capacitance values of the first and third shunt capacitors to adjust currents of the first and fourth areas.

A second aspect of the present disclosure provides a plasma processing apparatus comprising: a chamber having a processing space defined therein for processing a substrate; a support unit disposed in the chamber and supporting the substrate thereon; a gas supply unit for supplying gas to the processing space; and a plasma generation unit configured to excite gas in the processing space into a plasma state, wherein the plasma generation unit includes: a first power supply to supply power for generation of the plasma; a coil connected to the first power supply; and first to third shunt

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capacitors respectively disposed between and connected to first to third nodes of the coil and a ground.

In one implementation of the second aspect, the coil is divided into first to fourth areas, wherein the first node is disposed between the first and second areas, the second node is disposed between the second and third areas, and the third node is disposed between the third and fourth areas, wherein the apparatus is configured to control capacitance values of the first to third shunt capacitors to adjust currents of the second to fourth areas.

In one implementation of the second aspect, the plasma generation unit, further includes a fourth capacitor disposed between and connected to a fourth node other than the third node of the coil and the ground, wherein the apparatus is configured to control a capacitance value of the fourth capacitor to adjust a voltage of the coil.

A third aspect of the present disclosure provides a plasma processing apparatus comprising: a chamber having a processing space defined therein in which plasma is generated; and a plasma generation unit configured to excite gas in the processing space into a plasma state, wherein the plasma generation unit includes: a first power supply to supply power for generation of the plasma; a first coil portion having one end connected to the first power supply, wherein the first coil portion is divided into first and second areas; a second coil portion disposed under the first coil portion, and connected to the other end of the first coil portion, wherein the second coil portion is divided into third and fourth areas; and first to third shunt capacitors and a fourth capacitor respectively disposed between and connected to the first to fourth areas and a ground.

In one implementation of the third aspect, each of the first area and the third area constitutes one side half of each of the first and second coil portions while the first area and the third area face vertically each other, wherein each of the second area and the fourth area constitutes the other side half of each of the first and second coil portions while the second area and the fourth area face vertically each other.

In one implementation of the third aspect, the apparatus is configured to control capacitance values of the first and third shunt capacitors to adjust currents of the first and fourth areas.

In one implementation of the third aspect, the apparatus is configured to adjust currents of the first and fourth areas to control a density of the plasma.

A fourth aspect of the present disclosure provides a plasma processing method using a plasma processing apparatus, wherein the apparatus includes: a chamber having a processing space defined therein for processing a substrate; a support unit disposed in the chamber and supporting the substrate thereon; a gas supply unit for supplying gas to the processing space; and a plasma generation unit configured to excite gas in the processing space into a plasma state, wherein the plasma generation unit includes: a first power supply to supply power for generation of the plasma; a coil connected to the first power supply; and first to third shunt capacitors respectively disposed between and connected to first to third nodes of the coil and a ground, wherein the plasma processing method includes controlling capacitance values of the first to third shunt capacitors to adjust a current of the coil.

In one implementation of the fourth aspect, the coil is divided into first to fourth partial coils, wherein the first node is disposed between the first and second partial coils, the second node is disposed between the second and third partial coils, and the third node is disposed between the third and fourth partial coils, wherein the plasma processing method

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includes: controlling a capacitance value of the first shunt capacitor to adjust currents of the first and second partial coils; controlling a capacitance value of the second shunt capacitor to adjust currents of the second and third partial coils; and controlling a capacitance value of the third shunt capacitor to adjust currents of the third and fourth partial coils.

In one implementation of the fourth aspect, the plasma generation unit further includes a fourth capacitor disposed between and connected to a fourth node other than the third node of the coil and the ground, wherein the plasma processing method further includes controlling a capacitance value of the fourth capacitor to adjust a voltage of the coil.

In one implementation of the fourth aspect, the plasma processing method includes adjusting a current of the coil to control a density of the plasma.

The specific details of other embodiments are included in the detailed description and drawings.

BRIEF DESCRIPTION OF DRAWINGS

The above and other aspects and features of the present disclosure will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings, in which:

FIG. 1 is an exemplary diagram for illustrating a plasma processing apparatus according to some embodiments of the present disclosure;

FIG. 2 is an exemplary diagram for illustrating an antenna of the plasma processing apparatus of FIG. 1;

FIG. 3 is a circuit diagram for illustrating an equivalent circuit of the antenna in FIG. 2;

FIG. 4 is an exemplary view showing an upper partial coil and a lower partial coil of the antenna of FIG. 2; and

FIG. 5 is a flowchart for illustrating a plasma processing method according to some embodiments of the present disclosure.

DETAILED DESCRIPTIONS

For simplicity and clarity of illustration, elements in the drawings are not necessarily drawn to scale. The same reference numbers in different drawings represent the same or similar elements, and as such perform similar functionality. Further, descriptions and details of well-known steps and elements are omitted for simplicity of the description. Furthermore, in the following detailed description of the present disclosure, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be understood that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the present disclosure. Examples of various embodiments are illustrated and described further below. It will be understood that the description herein is not intended to limit the claims to the specific embodiments described. On the contrary, it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the present disclosure as defined by the appended claims.

A shape, a size, a ratio, an angle, a number, etc. disclosed in the drawings for illustrating embodiments of the present disclosure are exemplary, and the present disclosure is not limited thereto. The same reference numerals refer to the same elements herein. Further, descriptions and details of well-known steps and elements are omitted for simplicity of

the description. Furthermore, in the following detailed description of the present disclosure, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. However, it will be understood that the present disclosure may be practiced without these specific details. In other instances, well-known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the present disclosure.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the present disclosure. As used herein, the singular forms “a” and “an” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprise”, “comprising”, “include”, and “including” when used in this specification, specify the presence of the stated features, integers, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, operations, elements, components, and/or portions thereof. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expression such as “at least one of” when preceding a list of elements may modify the entirety of list of elements and may not modify the individual elements of the list. When referring to “C to D”, this means C inclusive to D inclusive unless otherwise specified.

It will be understood that, although the terms “first”, “second”, “third”, and so on may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section described below could be termed a second element, component, region, layer or section, without departing from the spirit and scope of the present disclosure.

In addition, it will also be understood that when a first element or layer is referred to as being present “on” or “beneath” a second element or layer, the first element may be disposed directly on or beneath the second element or may be disposed indirectly on or beneath the second element with a third element or layer being disposed between the first and second elements or layers. It will be understood that when an element or layer is referred to as being “connected to”, or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer, or one or more intervening elements or layers may be present. In addition, it will also be understood that when an element or layer is referred to as being “between” two elements or layers, it may be the only element or layer between the two elements or layers, or one or more intervening elements or layers may also be present.

Further, as used herein, when a layer, film, region, plate, or the like may be disposed “on” or “on a top” of another layer, film, region, plate, or the like, the former may directly contact the latter or still another layer, film, region, plate, or the like may be disposed between the former and the latter. As used herein, when a layer, film, region, plate, or the like is directly disposed “on” or “on a top” of another layer, film, region, plate, or the like, the former directly contacts the latter and still another layer, film, region, plate, or the like is not disposed between the former and the latter. Further, as used herein, when a layer, film, region, plate, or the like may be disposed “below” or “under” another layer, film, region, plate, or the like, the former may directly contact the latter

or still another layer, film, region, plate, or the like may be disposed between the former and the latter. As used herein, when a layer, film, region, plate, or the like is directly disposed “below” or “under” another layer, film, region, plate, or the like, the former directly contacts the latter and still another layer, film, region, plate, or the like is not disposed between the former and the latter.

Unless otherwise defined, all terms including technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this inventive concept belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

In one example, when a certain embodiment may be implemented differently, a function or operation specified in a specific block may occur in a sequence different from that specified in a flowchart. For example, two consecutive blocks may be actually executed at the same time. Depending on a related function or operation, the blocks may be executed in a reverse sequence.

In descriptions of temporal relationships, for example, temporal precedent relationships between two events such as “after”, “subsequent to”, “before”, etc., another event may occur therebetween unless “directly after”, “directly subsequent” or “directly before” is not indicated. The features of the various embodiments of the present disclosure may be partially or entirely combined with each other, and may be technically associated with each other or operate with each other. The embodiments may be implemented independently of each other and may be implemented together in an association relationship. Spatially relative terms, such as “beneath”, “below”, “lower”, “under”, “above”, “upper”, and the like, may be used herein for ease of explanation to describe one element or feature’s relationship to another element or feature as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or in operation, in addition to the orientation depicted in the figures. For example, when the device in the drawings may be turned over, elements described as “below” or “beneath” or “under” other elements or features would then be oriented “above” the other elements or features. Thus, the example terms “below” and “under” may encompass both an orientation of above and below. The device may be otherwise oriented, for example, rotated 90 degrees or at other orientations, and the spatially relative descriptors used herein should be interpreted accordingly.

FIG. 1 is an exemplary diagram for illustrating a plasma processing apparatus according to some embodiments of the present disclosure. FIG. 2 is an exemplary diagram for illustrating an antenna of the plasma processing apparatus of FIG. 1. FIG. 3 is a circuit diagram for illustrating an equivalent circuit of the antenna in FIG. 2. FIG. 4 is an exemplary view showing an upper partial coil and a lower partial coil of the antenna of FIG. 2.

First, referring to FIGS. 2 and 3, a plasma generation unit 400 according to some embodiments includes a first power supply 10, a coil 410, first to third shunt capacitors C1, C2, and C3, and a fourth capacitor C4.

The first power supply 10 supplies power for plasma generation. The first power supply 10 may supply high-frequency power.

The coil **410** may be connected to the first power supply **10** to generate plasma in a chamber **100**. The coil **410** may be embodied as a coil-type antenna for generating plasma.

Referring to FIG. 3, the coil **410** may be divided into a plurality of portions based on a position. The coil **410** may have a first node **P1** receiving an RF signal. Specifically, the coil **410** may be divided into first, second, third, and fourth partial coils **411**, **412**, **413** and **414**. In this regard, the first and second partial coils **411** and **412** are connected in series to each other while the first node **P1** is interposed therebetween. The second and third partial coils **412** and **413** are connected in series to each other while a second node **P2** is interposed therebetween. The third and fourth partial coils **413** and **414** are connected in series to each other while a third node **P3** is interposed therebetween.

In some embodiments, the first to fourth partial coils **411**, **412**, **413**, and **414** of the coil **410** may be referred to as first to fourth areas, respectively.

The first to third shunt capacitors **C1**, **C2**, and **C3** and the fourth capacitor **C4** are connected to the first to fourth partial coils **411**, **412**, **413**, and **414** of the coil **410**, respectively.

Specifically, the first shunt capacitor **C1** is connected to and disposed between the first node **P1** of the coil **410** and a ground. The second shunt capacitor **C2** is connected to and disposed between the second node **P2** other than the first node **P1** and the ground. The third shunt capacitor **C3** is connected to and disposed between the third node **P3** other than the second node **P2** and the ground. The fourth capacitor **C4** is connected to and disposed between and a fourth node **P4** other than the third node **P3** and the ground.

In some embodiments, each of the first to third shunt capacitors **C1**, **C2**, and **C3** may be embodied as a variable capacitor. The fourth capacitor **C4** may act as a balance capacitor.

Specifically, currents of the first and second partial coils **411** and **412** may be adjusted by controlling a capacitance value of the first shunt capacitor **C1**. Currents of the second and third partial coils **412** and **413** may be adjusted by controlling a capacitance value of the second shunt capacitor **C2**. Currents of the third and fourth partial coils **413** and **414** may be adjusted by controlling a capacitance value of the third shunt capacitor **C3**.

The fourth capacitor **C4** may play a role in adjusting a voltage of the coil **410**. When the fourth capacitor **C4** has a specific capacitance value, the voltage of the coil **410** may be adjusted by making a voltage at a center of the coil **410** become 0V. In this case, the center of the coil **410** may refer to a point at which a voltage of a node of the second shunt capacitor **C2** becomes 0V, and may mean a point at which a sign of the voltage of the coil **410** changes. The center of the coil **410** may mean a point between the first area **411** and the third area **413** arranged at a left side and the second area **412** and the fourth area **414** arranged at a right side around a imaginary line **L** of FIG. 4.

Referring to FIG. 2 and FIG. 4, the coil **410** includes a first coil portion **400a** having an end connected to the first power supply **10** and divided into the first and second areas **411** and **412**, and a second coil portion **400b** disposed below the first coil portion **400a** and connected to the other end of the first coil portion **400a** and divided into the third and fourth area **413** and **414**. In this case, the first coil portion **400a** may mean an upper coil, while the second coil portion **400b** may mean a lower coil.

Each of the first area **411** and the third area **413** may constitute one half of each of the first and second coils **400a** and **400b** and may face each other. Each of the second area **412** and the fourth area **414** may constitute the other half of

each of the first and second coils **400a** and **400b** and may face each other. In this case, each of the first area **411** and the third area **413** may constitute a left half of each of the first and second coils **400a** and **400b** around the imaginary line **L** in FIG. 4. Each of the second area **412** and the fourth area **414** may constitute a right half of each of the first and second coils **400a** and **400b** around the imaginary line **L** in FIG. 4. A location of the imaginary line **L** is not limited to that as shown in FIG. 4.

Referring to Tables 1 and 2 below, a magnitude of the current of each of the first to fourth areas **411**, **412**, **413**, and **414** may vary depending on each of the capacitance values of the first to third shunt capacitors **C1**, **C2**, and **C3** and the fourth capacitor **C4**.

In other words, the magnitude of the current of each of the first to third shunt capacitors **C1**, **C2**, and **C3** and the fourth capacitor **C4** as measured via fifth to seventh ammeters **A5**, **A6**, and **A7** may vary based on each of the capacitance values of the first to third shunt capacitors **C1**, **C2**, and **C3** and the fourth capacitor **C4**. As the magnitude of the current of each of the first to third shunt capacitors **C1**, **C2**, and **C3** varies, the magnitude of the current of each of the first to fourth areas **411**, **412**, **413**, **414** of the coil **410** as measured via first to fourth ammeters **A1**, **A2**, **A3**, and **A4** may vary.

Specifically, as the capacitance of each of the first to third shunt capacitors **C1**, **C2**, and **C3** increases, the magnitude of the current of each of the first to third shunt capacitors **C1**, **C2**, and **C3** may increase, so that the magnitude of the current of the coil **410** may decrease.

On the contrary, as the capacitance of each of the first to third shunt capacitors **C1**, **C2**, and **C3** decreases, the magnitude of the current of each of the first to third shunt capacitors **C1**, **C2**, and **C3** may decrease, so that the magnitude of the current of the coil **410** may increase.

For example, referring to Table 1, it may be identified that in Present Example 1, the capacitance value of the first shunt capacitor **C1** increases compared to Comparative Example. Referring to Table 2, it may be identified that as the magnitude of the current of the first shunt capacitor **C1** as measured via the fifth ammeter **A5** increases compared to the Comparative Example, the magnitude of the current of the first area **411** as measured via the first ammeter **A1** decreases compared to the Comparative Example, and the magnitude of the current of the second area **412** measured via the second ammeter **A2** increases compared to the Comparative Example.

In Present Example 2, as the capacitance value of the first shunt capacitor **C1** increases compared to Present Example 1, the magnitude of the current of the first area **411** may decrease compared to Present Example 1. In this case, the magnitude of the current of the first area **411** may be adjusted to be smaller than the magnitude of the current of each of the second to fourth areas **412**, **413**, and **414**, so that the plasma density of the first area **411** may be reduced.

It may be identified that, in each of Present Example 3 and Present Example 4, the capacitance value of the second shunt capacitor **C2** increases compared to the Comparative Example. This may indicate that each of voltages of the first area **411** and the third area **413** disposed at the left side around the imaginary line and each of the voltages of the second area **412** and the fourth area **414** disposed at the right around the line are balanced with each other. In this case, it may be identified that there is little change in the magnitude of the current based on the capacitance value of the second shunt capacitor **C2**.

It may be identified that in each of Present Example 5 and Present Example 6, the capacitance value of the third shunt

In the Present Example 2, the sum of the magnitudes of the currents of the first and third areas **411** and **413** constituting the left area of the coil **410** is 1183 mA, while the sum of the magnitudes of the currents of the second and fourth

That is, adjusting the capacitance values of the first to third shunt capacitors C1, C2, and C3 may allow the magnitude of the current of the coil 410 to be freely controlled based on each of the areas of the coil 410. As a result, the plasma density of the coil 410 may be freely controlled based on each of the areas of the coil 410.

[illegible]

TABLE 2

	Comparative Example	Present Example 1	Present Example 2	Present Example 3	Present Example 4	Present Example 5	Present Example 6
First ammeter A1	581 mA	564 mA	472 mA	581 mA	581 mA	567 mA	522 mA
Second ammeter A2	585 mA	605 mA	711 mA	585 mA	585 mA	570 mA	526 mA
Third ammeter A3	585 mA	605 mA	711 mA	585 mA	586 mA	571 mA	527 mA
Fourth ammeter A4	581 mA	601 mA	707 mA	581 mA	582 mA	535 mA	395 mA
Fifth ammeter A5	4 mA	40 mA	239 mA	4 mA	4 mA	4 mA	4 mA
Sixth ammeter A6	0	0	0	0	2 mA	0	1 mA
Seventh ammeter A7	4 mA	4 mA	5 mA	4 mA	4 mA	36 mA	132 mA

FIG. 5 is a flowchart for illustrating a plasma processing method according to some embodiments of the present disclosure. For convenience of description, a following description is mainly based on differences from the descriptions using FIG. 1 to FIG. 4.

A plasma processing apparatus according to some embodiments includes a chamber 100, a support unit 200, a gas supply unit 300 and the plasma generation unit 400.

The plasma generation unit 400 includes the first power supply 10 that supplies the power for plasma generation, the coil 410 connected to the first power supply 10, and the first to third shunt capacitors C1, C2, and C3 respectively connected to and disposed between the first to third nodes P1, P2, and P3 and the ground.

In the plasma processing apparatus according to some embodiments, the current of the coil 410 may be adjusted by controlling the capacitance values of the first to third shunt capacitors C1, C2, and C3.

Referring to FIG. 3 and FIG. 5, the currents of the first and second partial coils 411 and 412 is adjusted by controlling the capacitance value of the first shunt capacitor C1 in S10.

The currents of the second and third partial coils 412 and 413 is adjusted by controlling the capacitance value of the second shunt capacitor C2 in S20.

The currents of the third and fourth partial coils 413 and 414 is adjusted by controlling the capacitance value of the third shunt capacitor C3 in S30.

The plasma generation unit 400 includes the fourth capacitor C4 connected to and disposed between the fourth node P4 other than the third node P3 and the ground. The voltage of the entirety of the coil 410 is adjusted by controlling the capacitance value of the fourth capacitor C4 in S40.

In some embodiments, each of the first to third shunt capacitors C1, C2, and C3 may be embodied as a variable capacitor, and the fourth capacitor C4 may be embodied as a balance capacitor.

That is, the capacitance values of the first to third shunt capacitors C1, C2, and C3 and the fourth capacitor C4 may be adjusted such that the magnitude of the current of the coil 410 may be freely controlled based on each of the areas of the coil 410. As a result, the plasma density of coil 410 may be freely controlled based on each of the areas of the coil 410.

FIG. 1 is an exemplary cross-sectional view for illustrating a plasma processing apparatus according to some embodiments of the present disclosure. FIG. 1 illustrates a plasma processing apparatus for generating plasma in an ICP (Inductively Coupled Plasma) scheme by way of example. However, the disclosure is not limited thereto.

Referring to FIG. 1, a plasma processing apparatus 1000 according to some embodiments of the present disclosure may act as a substrate processing apparatus that processes a substrate W using plasma. For example, the plasma process-

ing apparatus 1000 may perform an etching process on the substrate W. The plasma processing apparatus 1000 may include a chamber 100, a support unit 200, a gas supply unit 300, and a plasma generation unit 400.

The chamber 100 provides a space where the substrate processing process is performed. The chamber 100 includes a housing 110 and a sealing cover 120.

The housing 110 has an inner space having an open top. The inner space of the housing 110 serves as a processing space where the substrate processing process is performed. The housing 110 may be made of a metal material. The housing 110 may be made of an aluminum material. The housing 110 may be grounded. An exhaust hole 102 is formed in a bottom face of the housing 110, so that reaction by-products generated in the process and gas staying in the inner space of the housing may be discharged to an outside. The inside of the housing 110 is decompressed to a predetermined pressure by the exhaust process.

The sealing cover 120 covers the open top of the housing 110. The sealing cover 120 is formed in a shape of a plate and seals the inner space of the housing 110. The sealing cover 120 may include a dielectric material window.

The support unit 200 is located inside the housing 110. The support unit 200 supports the substrate W thereon. The support unit 200 may include an electrostatic chuck that adsorbs the substrate W using electrostatic force. Alternatively, the support unit 200 may support the substrate W in various schemes such as mechanical clamping.

The support unit 200 may be positioned inside the chamber 100 and may be spaced upwardly from a bottom face of the housing 110.

The gas supply unit 300 supplies a process gas into the chamber 100. The gas supply unit 300 includes a gas supply nozzle 310, a gas supply line 320 and gas storage 330. The gas supply nozzle 310 is installed in a center of the sealing cover 120. An injection hole is formed in a bottom of the gas supply nozzle 310. The injection hole is located under the sealing cover 120 and supplies the process gas to the processing space inside the chamber 100. The gas supply line 320 connects the gas supply nozzle 310 to the gas storage 330. The gas supply line 320 supplies the process gas stored in the gas storage 330 to the gas supply nozzle 310. A valve 321 is installed on the gas supply line 320. The valve 321 opens and closes the gas supply line 320, and adjusts a flow rate of the process gas to be supplied via the gas supply line 320.

The plasma generation unit 400 excites the process gas in the chamber 100 into a plasma state. According to one embodiment of the present disclosure, the plasma generation unit 400 may be of an ICP type.

The plasma generation unit 400 may include a high-frequency power supply device 420, the antenna 410, and a

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power distributor **430**. The high-frequency power supply device **420** supplies a high-frequency signal, that is, an RF signal. The high frequency power supply device **420** may include the first power supply **10** as described using FIG. 1 to FIG. 4. Although not specifically illustrated, the high-frequency power supply device **420** may further include an impedance matching system.

The antenna **410** is connected in series with the high frequency power supply device **420**. The antenna **410** may be embodied as a coil having multiple turns. The antenna **410** is electrically connected to the high frequency power supply device **420** and receives a RF power therefrom. The power distributor **430** distributes the power supplied from the high frequency power supply device **420** to the antenna **410**.

The antenna **410** may face the substrate **W**. For example, the antenna **410** may be installed on a top face of the chamber **100**. The antenna **410** may be formed in a ring shape. In this regard, the antenna **410** may be embodied as a plurality of rings having different radii.

According to an embodiment, the antenna **410** may be disposed on a top face of the chamber **100**, or may be disposed on a side face of the chamber **100**. As long as the antenna **410** generates the plasma within the chamber **100**, a position of the coil is not limited to a specific position.

The antenna **410** may receive the RF power from the high-frequency power supply device **420** to induce a time-varying electromagnetic field in the chamber. Accordingly, the process gas supplied to the chamber **100** may be excited to the plasma.

Although the embodiments of the present disclosure have been described with reference to the accompanying drawings, a person with ordinary skill in the technical field to which the present disclosure belongs will appreciate that the present disclosure may be implemented in various forms without changing technical idea or essential features thereof. Therefore, it should be understood that the embodiments as described above are illustrative in all respects and not restrictive.

What is claimed is:

1. A plasma processing apparatus comprising:

a chamber having a processing space defined therein in which plasma is generated; and

a plasma generation unit configured to excite gas in the processing space into a plasma state,

wherein the plasma generation unit includes:

a first power supply to supply power for generation of the plasma;

a coil connected to the first power supply;

a first shunt capacitor connected between a first node of the coil and a ground; and

a second shunt capacitor connected between a second node other than the first node of the coil and the ground, wherein:

the coil comprises a first partial coil and a second partial coil connected in series to each other,

one end of the first partial coil is connected to the first power supply and another end of the first partial coil is connected to the first node of the coil connected to the first shunt capacitor, and

one end of the second partial coil is connected to the first node of the coil connected to the first shunt capacitor and another end of the second partial coil is connected to the second node of the coil connected to the second shunt capacitor.

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2. The plasma processing apparatus of claim 1, wherein the apparatus is configured to control a capacitance value of each of the first and second shunt capacitors to adjust a current of the coil.

3. The plasma processing apparatus of claim 1, wherein the apparatus is configured to adjust a current of the coil to control a density of the plasma.

4. The plasma processing apparatus of claim 1, wherein each of the first and second shunt capacitors includes a variable capacitor.

5. The plasma processing apparatus of claim 1, wherein the first node is interposed between the first partial coil and the second partial coil,

wherein the apparatus is configured to control a capacitance value of the first shunt capacitor to adjust currents of the first and second partial coils.

6. The plasma processing apparatus of claim 5, wherein the coil is further divided into a third partial coil connected in series with the second partial coil while the second node is interposed therebetween,

wherein the apparatus is configured to control a capacitance value of the second shunt capacitor to adjust currents of the second and third partial coils.

7. The plasma processing apparatus of claim 6, wherein the plasma generation unit further includes a third shunt capacitor disposed between and connected to a third node other than the second node of the coil and the ground,

wherein the coil is further divided into a fourth partial coil connected in series with the third partial coil while the third node is interposed therebetween,

wherein the apparatus is configured to control a capacitance value of the third shunt capacitor to adjust currents of the third and fourth partial coils.

8. The plasma processing apparatus of claim 7, wherein the plasma generation unit further includes a fourth capacitor disposed between and connected to a fourth node other than the third node of the coil and the ground,

wherein the apparatus is configured to control a capacitance value of the fourth capacitor to adjust a voltage of the coil.

9. The plasma processing apparatus of claim 1, wherein the coil includes:

a first coil portion having one end connected to the first power supply, wherein the first coil portion is divided into first and second areas; and

a second coil portion disposed below the first coil portion and connected to the other end of the first coil portion, wherein the second coil portion is divided into third and fourth areas,

wherein each of the first area and the third area constitutes one side half of each of the first and second coil portions while the first area and the third area face vertically each other,

wherein each of the second area and the fourth area constitutes the other side half of each of the first and second coil portions while the second area and the fourth area face vertically each other,

wherein the apparatus is configured to control capacitance values of the first and third shunt capacitors to adjust currents of the first and fourth areas.

10. A plasma processing apparatus comprising:

a chamber having a processing space defined therein in which plasma is generated; and

a plasma generation unit configured to excite gas in the processing space into a plasma state,

wherein the plasma generation unit includes:

a first power supply to supply power for generation of the plasma;

a first coil portion having one end connected to the first power supply, wherein the first coil portion is divided 5
into first and second areas;

a second coil portion disposed under the first coil portion, and connected to the other end of the first coil portion, wherein the second coil portion is divided into third and fourth areas; and 10

first to third shunt capacitors and a fourth capacitor connected between the first to fourth areas and a ground, respectively.

11. The plasma processing apparatus of claim **10**, wherein each of the first area and the third area constitutes one side 15
half of each of the first and second coil portions while the first area and the third area face vertically each other,

wherein each of the second area and the fourth area constitutes the other side half of each of the first and second coil portions while the second area and the 20
fourth area face vertically each other.

12. The plasma processing apparatus of claim **10**, wherein the apparatus is configured to control capacitance values of the first and third shunt capacitors to adjust currents of the first and fourth areas. 25

13. The plasma processing apparatus of claim **12**, wherein the apparatus is configured to adjust currents of the first and fourth areas to control a density of the plasma.

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